

Protocol for eligibility rating

- 1) The eligibility rating will be performed by two independent raters. Discrepancies will be discussed until consensus is found. To examine interrater reliability, Cohen's kappa κ will be calculated for the initial ratings.
- 2) The randomly-numbered studies will be screened starting with the result randomly matched with number one. Screening will be continued until a minimum of $n=60$ studies are found to be eligible, i.e., meeting all inclusion criteria.

To be included into the ReFiNe-MDD project, the original study has to match the following criteria:

Checklist



- The publication must contain primary research (no systematic reviews or meta-analyses)
- The publication must be written either in German or English.
- The publication must be published in a peer-reviewed journal.
- The publication must address a neurobiological finding associating some kind of depression-related variable (described below) with gray matter brain structure.
 - Examples for depression-related variables are: risk factors (e.g., genetic or environmental risk), protective factors, diagnosis, symptoms, course of disease, subtypes of the disorder or similar variables.
- Gray matter structure must be measured with a 3T field strength scanner producing T1-weighted scans.
- A voxel-based morphometry approach (VBM; minimum voxel resolution 1.5 mm) must be used.
- Voxel must be approximately isotropic (e.g., $1.2 \times 1.2 \times 1.5$ would be okay, $1.2 \times 1.2 \times 2.5$ would not be okay).
- The publication must include at least one significant effect regarding gray matter volume.
- For analysis, standard univariate statistical methods must be utilized.
- The sample and methods included in the original study have to be sufficiently comparable to samples and methods available within the replication data (see [Description of Subsamples](#) and [Overview of instruments](#)). The comparability should be rated based on the specific research question. Examples for comparable and non-comparable cases are defined here [Comparability examples](#).
 - Sample characteristics: Comparability should be checked for example regarding age, somatic disease status, mental health disease status, medication status and other clinical characteristics.
 - Methods: Comparability of utilized methods should be checked for example regarding the utilized instrument and operationalization measuring and defining the predictor variable.
 - The replication sample with comparable sample characteristics and comparable methods must have at least 90 % of the sample size of the original sample.

- 3) In case, a study is included raters have to check whether there is only one or multiple reported statistic findings. If there are multiple findings, the key finding has to be identified. This task is performed independently by two raters as well. Again, discrepancies will be discussed until consensus is found; To examine interrater reliability, Cohen's kappa κ will be calculated for the initial ratings.
- 4) The list of eligible studies for replications is checked for duplicates. In case studies have to be dropped due to being a duplicate, the list of all studies will be consulted again to fill up the number of 60 studies to replicate.